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Spirituality, Religiosity and Addiction Recovery: Current Perspectives



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Abstract: Substance use disorders are an important public health problem with a multifactorial etiology and limited effective treatment options. Within this context, spirituality-based approaches may provide interesting and useful options in managing substance use disorders. This kind of intervention can have positive effects in alleviating some core symptoms associated with substance use, such as aggressiveness. Improvement in cessation rates for alcohol, cocaine and opioid use disorders have also been described in some clinical studies. However, spirituality may not play a beneficial role in some subgroups, such as among individuals with crack cocaine and cannabis use disorders. A widely available intervention for alcohol use disorders is Alcoholics Anonymous (AA), which can be seen as a spirituality-based intervention. Spirituality also seems to be especially beneficial for minorities such as Latinos, African-Americans and Native-Americans. Moreover, spiritual-based interventions are also helpful alternatives in many rural environments where conventional healthcare for substance use disorders may not be easily available. However, spiritual-based interventions may be considered as a possible adjunctive therapeutic option to conventional treatments. There is a need for prospective studies outside U.S., especially where spiritual-based approaches are available. It may be difficult to carry out randomized controlled trials because of the nature of the spiritual/religious dimensions. However, prospective studies that evaluate mediation effect of spirituality and religiosity on recovery would be helpful. Qualitative studies combined with quantitative design offer excellent options to evaluate the recovery process, especially among special populations.

ARTICLE HISTORY

Received: August 29, 2017
Revised: May 31, 2018
Accepted: June 04, 2018

DOI:
10.2174/1874473711666180612075954



CrossMark

Keywords: Spirituality, religiosity, alcohol, cocaine, opioid, alcoholics anonymous.

1. INTRODUCTION

In the most recent report from the Global Burden of Disease Study, Substance Use Disorders (SUD) directly accounted for 85 (alcohol) and 20 (other drugs) million disability-adjusted life-years [1, 2]. Thus, substance use disorders are an important public health issue across the globe. Vulnerability for SUDs can be understood as the result of various factors which may interact (*e.g.*, genetics, personality and environment). Among environmental factors, spirituality plays an important role [3]. Considering that SUD recovery is difficult to achieve, it is important to explore all potential factors involved.

Spirituality has been of interest for managing individuals on alcohol-dependence following the 12-step group therapy developed in 1935, and those performed by Roman Catholic

groups [4-10]. The association between spirituality and substance dependence has been one of the best-documented protective relationships in the literature [11, 12]. It has been also shown that spirituality may help those individuals with SUDs to manage relational socio-cultural factors such as re-socialization and family relationships [13, 14]. Spirituality can be helpful to cope with specific anguish and needs among alcohol abusers [13, 14]. It has been observed that daily spiritual experiences and sense of purpose/meaning in life are associated with alcohol abstinence at 6 months as well as participation to the Alcoholics Anonymous (AA) [15]. This finding suggests that spirituality and religiosity can positively influence treatment outcome in alcohol dependence [16]. Spirituality can also be defined as something that gives meaning and purpose to life [17] as well as a sense of personal identity and transcendence that motivates the individual beyond the everyday issues of life [18]. Religiosity, on the other hand, is considered as the understanding of a particular system or dogma based on faith and participation in the rituals and services offered by a community united by faith [19].

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Spirituality and SUDs have coexisted in the world for millennia [20]. Substances of different kinds had been used to reach trance levels. Shamans and other religious healers have often used substances to bring about spiritual experiences [21]. Similarly, spiritual interventions have been affecting the treatment outcomes for patients with SUDs [22]. For example, spirituality is a core concept in Alcoholics Anonymous (AA), being included in most of the 12-steps taken by members enrolled in the program, as well as in the 12 traditions of AA groups [23].

The present narrative review aims to summarize the evidences on the impact of spirituality and religiosity on the recovery rates of individuals with SUDs. We reviewed studies on the impact of spirituality and religiosity among patients with SUDs and the role of AA groups in the recovery process worldwide. Moreover, we also reviewed some studies on specific populations such as Latinos, African Americans, Muslims and Rural groups, which traditionally have different patterns for addictive behaviors. Table 1 summarizes the findings and limitations of the main studies included in the present review.

2. SPIRITUALITY AND RECOVERY

Local culture and spirituality [20, 24-27] have been shown to be influencing factors in the recovery of individuals with SUDs. Historically, drugs have been used in religious ceremonies, by healers for medicinal purposes, or by the general population in a socially approved way [20]. Among university students, drug use was negatively associated with religiosity [27]. However, more studies are needed to identify the mechanisms by which religiosity exerts this protective influence. In fact, they may be part of those motivational and behavioral factors (intrinsic and extrinsic) which may lead to a change in life of people with SUDs [28, 29]. Also, these variables can influence motivation and readiness for changing along with personal values, decision-making, way of living [30].

2.1. Aggressiveness

Individuals with SUDs are more likely to be violent and show aggressive behavior [31, 32]. Part of this risk is due to the effects of substances on information processing capabilities and reduced ability to respond to conflicting situations in a non-aggressive way [33, 34]. Moreover, recent evidences suggest that self-control functions can mediate the influence of religiousness on substance use [35]. In fact, it is considered that aggressive behavior occurs in part because of the lack of self-control [36]. Despite decreasing with successful substance use treatment, many individuals with SUD continue with aggressive behavior following treatment. Considering that many substance use treatment programs employ spirituality-based interventions, Shorey *et al.* [32] examined the relationship between spirituality and aggression among men in residential treatment for substance use. Attitudinal, physical, and verbal aggression, as well as a composite aggression score were associated with higher spirituality. After controlling for age, alcohol use and problems, and drug use and problems, spirituality remained negatively associated with aggression.

2.2. Culture

For a long time, alcohol/drug use was considered a desirable behavior, and an important aspect for socialization and social activities in many cultures [20]. Although the harmful effects of alcohol use are well documented [37], its rate of use/abuse remains high worldwide [38]. However, not surprisingly it may vary widely on the base of cultural values including religious attitudes to alcohol and drinking [39]. Norrish & Jooste [14] confirmed that socio-cultural aspects are essential in promoting patient re-socialization by encouraging contact with family and friends, as well as association with others.

Acculturation is the process of cultural change resulting from the contact of individuals from different cultures [40]. Behaviors and attitudes from the original culture may change according to the new environmental factors. In fact, immigrants exposed to new values and situations can adopt the practices and traditions of the dominant culture in the new country. Such changes may lead to increased use of alcohol and other substances [41]. This may be attributed to foster a sense of belonging, feeling relaxed and also to deal with loneliness, *etc.* Some individuals may feel acculturated whereas others may maintain their traditions in an extreme way [42].

Spirituality does influence alcohol consumption behaviors, but there is still a lot that needs to be explored in this field [43]. A recent systematic review found that alcohol first use and alcohol use disorders are influenced by religion and values of local culture are associated with alcohol use. Moreover, religion and culture have an opposite role on substance use and misuse in Western countries: religion is a protective factor for minorities whereas a hard-drinking culture seems to be the risk factor [43].

2.3. Clinical Studies

Although changes in spirituality and spiritual behaviors are hypothesized to have a role in the recovery process from SUDs, few prospective studies have been conducted in this field. Robinson *et al.* [16] recruited 364 individuals with alcohol dependence from among patients belonging to treatment centers based on abstinence, patients in moderate alcohol consumption programs and individuals who were not receiving treatment. The aim was to study the long-term effects of spirituality/religion among these patients on the base of their "AA involvement". Spirituality and religion were divided into 12 categories: Loving God Scale; Controlling God Scale; Meaning/values/beliefs; Positive religious coping; Beliefs; Private Spirituality/Religiousness practice; Daily spiritual experiences; Fetter forgiveness; Mauger forgiveness of self; Mauger forgiveness of others; Negative religious coping; and Purpose in life. The study found a significant reduction in alcohol consumption over 6 months in 8 of the 12 categories (the 8 last listed above). Authors concluded that changes in spirituality/religiosity had a positive impact on alcohol consumption, even when controlled for the variable "AA involvement", and spirituality/religiosity deserves attention for generating positive change even among individuals not affiliated of AA or religious institutions [16]. Well-designed prospective studies have shown that spirituality/religiosity can improve some markers of physical

Table 1. Findings of the main studies included in the present narrative review.

Author	Year	Type	Section	Main Findings	Limitations
De Wall <i>et al.</i> [35]	2014	Prospective	2.1	Self-control functions as a mediator of the relationship between various indicators of religiousness and substance use	The size of its mediating effect was small. Self-report alcohol and drug use
Castaldelli-Maia & Bhugra [43]	2014	Systematic Review	2.2	Religion and culture seem to have an opposite role on substance use and misuse in Western countries: religion being a protective factor while culture a risk factor	Only included English-language articles. Search in just one database
Robinson <i>et al.</i> [15]	2007	Prospective	2.3	Changes in spirituality/religiosity had a positive impact on alcohol consumption even when controlled AA involvement, even in individuals not affiliated with AA or religious institutions.	Just one clinical site sample. All the individuals were receiving health professional support
Heinz <i>et al.</i> [49]	2007	Prospective	2.3	Religious/spiritual experiences facilitated the development of positive attitudes towards stressful situations and better coping without relapse	Atypical outpatient sample, in which only 11% had ever been involved in 12-step programs
Kelly <i>et al.</i> [23]	2016	Review	3	AA's ability in: helping to establish social networks to support abstinence and recovery; achieving abstinence through self-efficacy and the development of coping skills; and supporting individuals in the maintenance of recovery motivation over time	Non-systematic review. At least 50% percent of the direct effects of AA was not explained
Garcia <i>et al.</i> [68]	2015	Ethnographic	4.1	Group High Power (GHP) provides empirical foundations on which it seeks to re-theorize the importance of spirituality and culture in mutual aid recovery groups without, however, defining the Latino population as homogenous	No outcome evaluation
Taylor <i>et al.</i> [71]	2003	Cross-sectional	4.2	African-American involvement in religious communities has helped such a population deal with the hostile circumstances associated with psychological and emotional suffering	No causal investigation (cross-sectional study)
Stone <i>et al.</i> [75]	2006	Cross-sectional	4.3	American Natives participation in traditional activities and traditional spirituality (two of the three components of enculturation) had significant positive effects on alcohol cessation. In general, older adults, women, and married adults were more likely to achieve alcohol abstinence	No causal investigation (cross-sectional study)
Shamsalina <i>et al.</i> [79]	2014	Qualitative	4.4	Spiritual-based psychotherapy plays a crucial role in the onset and maintenance of recovery for Muslims	Non-successful cases were not included. Individuals just from Iran.

well-being [44]. However, cause-effect relationship needs to be demonstrated through prospective long-term studies. Several retrospective studies [45, 46] have supported the evidence that spirituality has a positive influence on patient recovery.

Twelve-step programs (as described below) are widely recognized as a spirituality-based approach and are based on a life-style change [47] associated with better treatment outcomes. These programs are also associated with higher rates of abstinence and an effective relapse prevention [48]. Heinz *et al.* [49] investigated the association between spiritual/religious experiences and clinical outcomes in a program for

heroin and cocaine dependence. They found that such experiences facilitated the development of positive attitudes towards stressful situations and better coping without any relapse [49].

Williamson and Hood Jr. [50] conducted one of the most interesting intervention studies in the field, the Lazarus project. It was a longitudinal study of spiritual transformation in a 12-month Pentecostal-Charismatic residency program for substance abuse. Williamson and Hood Jr. [50] administered to residents an assessment protocol consisting of psychological (depression, self-esteem, psychopathology, Big 5 personality markers) and religiosity (fundamentalism, religious

orientation, spiritual well-being, mysticism) measures at: induction; six months; graduation; and one-year post-graduation. They also assessed a membership group from the sponsoring church using the same protocol. They found that general change in graduate scores occurred from induction to six-month assessment and persisted to graduation and one-year post-graduation [50]. Graduate score patterns generally agreed with those of dropouts at T1, but diverged at T2, becoming more similar to score patterns of church members. A multivariate analysis found that less openness (a personality marker) was the most important predictor of dropout [50].

2.3.1. Alcohol Use Disorders

Specifically, religion and spirituality seem to be an important factor in managing disorders of alcohol use [43]. However, not surprisingly, this influence is not equally significant for all sub-populations. Even when there is a strong local culture of alcohol consumption, a return to the spiritual- or religious-based traditional culture for minorities positively influences the outcomes of alcohol use abusers. Spirituality and/or religion generate support for abstinence in the general population also. Castaldelli-Maia & Bhugra [43] suggest that it is possible to offer spiritual-based interventions as adjunctive therapies for people with alcohol use disorders. However, those with a religious behavior history, would experience a higher beneficial effect than those who not.

2.3.2. Cocaine/Crack

Schoenthaler *et al.* [51] used the NIDA Drug Addiction Treatment Outcome Study data set to examine relapse rates 12 months after broken down by five spirituality measures. Stronger spiritual/religious beliefs and practices were directly associated with remission from abused drugs except crack. Much like the value of having a sponsor, for clients who abuse drugs, regular spiritual practice, and particularly weekly attendance at the religious services of their choice is associated with significantly higher remission [51]. These results demonstrate the clinically significant role of spirituality and the social bonds it creates in drug treatment programs.

2.3.3. Cannabis Use disorders

Yaterian *et al.* [52] examined spirituality and religiosity as outcome predictors in an outpatient treatment adolescent sample in which cannabis was the predominant drug of choice. Qualitative data were used to contextualize the quantitative findings. They found that higher post-treatment levels of spirituality were associated with increased cannabis use at 6-month follow-up. However, religiosity did not predict substance use outcomes at later timepoints. In the qualitative phase, adolescents described believing that they had a choice about their substance use and were in control of it, feeling more spiritual when under the influence of cannabis, and being helped by substance use [52].

2.3.4. Opioid Use Disorders

Noormohammadi *et al.* [53] conducted a cross-sectional study with 312 Iranian addicted patients. The patients had at least an attempt of detoxification in the past six months and referred to an outpatient detoxification clinic. Paloutzian and

Ellison's Spiritual Well-being Scale, demographic characteristics, and questions about associated factors with relapse was administered. Those who had significantly lower scores of spiritual well-being and its subscales were associated with relapse when compared to those with higher scores [53].

3. TWELVE-STEP PROGRAMS

The search for knowledge and personal identity can be expressed through participation in some religious activity. However, this can go beyond and include involvement in family, community or arts [54]. AA is a movement that aims the recovery of the individual based on spirituality, considering substance use disorders as a disease of the soul, body and spirit. It requires a commitment to abstinence and includes 12 steps that take the individual into a new spiritual awakening or a new state of consciousness [55]. Research has identified several concepts through which spirituality expressed in 12-step may promote patients' recovery. Green *et al.* [56] showed that spirituality assists in improving social support by sharing beliefs and generating a feeling of belonging to a group [57]. The beliefs and spiritual practices associated with recovery may lead to greater cognitive vigilance against relapse, activating a "recovery scheme" that may lead to greater motivation and coping [58]. As such, it may also lead to a reduction of negative beliefs and depressive reaction to hostile life events [58]. In a systematic review that evaluated behavioral change mechanisms through which AA provided benefits, it was found that AA are helpful in: helping to establish social networks supporting abstinence and recovery; achieving abstinence through self-efficacy and the development of coping skills; supporting individuals in the maintenance of recovery motivation over time [22, 23].

AA is based on multiple areas of approach, not only spiritual concepts, including the medical need to abstain completely in order to avoid craving and compulsive use, including behavioral psychology, and group dynamics [58, 59]. In addition, biological factors involving various addictive behaviors (gambling, substance use, compulsive sexual behaviors, among others) characterized by loss of control cannot be ignored. Among these addictive behaviors, SUDs have been identified as the most associated with neuro-anatomical and neurochemical alterations, mainly involving the dopaminergic system [60]. Such neurophysiological changes associated with the use of substances may explain behaviors patterns and emotions among these individuals [61]. The relationship between spirituality and neurobiological aspects of recovery requires further exploration [62-64]. In addition, there are positive and negative aspects of Christianity within the 12-steps so it will be interesting to explore these through other cultures. Religious experiences associated with moral and stigmatizing judgments may play a role in shaping individual attitudes toward spiritual approaches [65]. Thus it will be of interest to see how AA approach would work in non-Christian cultures, such as Buddhist, Islamic or other [66].

4. SPECIAL POPULATIONS

4.1. Latinos

There is a group known as Group High Power (GHP) - a mutual aid group located in Northern California for the La-

tino population with SUDs – which is a culturally adapted form of the 12-step program. It provides empirical foundations for the importance of spirituality and culture in mutual-aid- recovery- groups without, however, defining the Latino population as homogenous. It differs from the AA groups because it does not aim only abstinence but also focuses on individuals' life. In GHP groups, the members can self-denominate as being an alcoholic, a drug addict, an emotional sick (neurotic), or any combination among these [67, 68]. One of the important features that really makes it a target for the Latino population is the fact that its practices resemble the charismatic pentangle and catholic movements which are very common in Mexico. Benish *et al.* [69] proposes that one of the key factors in psychotherapeutic adherence is the inclusion of ethnic minorities themselves. Another relevant point is the role of family involvement because its participation is not restricted to the patients [68].

4.2. African-American

Religion and spirituality play central role in the African-descendant community [70], in which both God and the priest figures are very important. Thus, involvement in religious communities has helped such population in dealing with the hostile circumstances associated with psychological and emotional suffering [71]. In addition, the constant participation of faith as a significant factor in reducing the use of substances over time, suggests that adherence to a religion can minimize uncertainties and fears about the recovery process [72], promoting a sense of well-being [73] and commitment to the improvement process [74].

4.3. Native-American

Stone *et al.* [75] used data from a 3-year lagged sequential study on four Native-American reservations in the U.S. and Canada. Respondents were guardians or parents of children aged between 10-12. This inclusion criterion was chosen since the project aimed to identify culturally specific resilience and risk factors that affect children well-being and development. They gathered data of a sample of 732 lifetime alcohol users, mainly women. According to prior researches, three basic enculturation dimensions, measures were developed in collaboration with participating tribes. Participation in traditional activities and traditional spirituality (two of the three components of enculturation) had significant positive effects on alcohol cessation. In general, older adults, women, and married adults were more likely to achieve alcohol abstinence [75].

4.4. Muslims

Iran is a middle-income country with culture in transition and thus is constantly changing at different economic and cultural levels. In this process, attention is drawn to the high prevalence of risk behaviors among young people, such as opioid dependence [76, 77]. Despite having a religious context and considerable level of support from family and society as protective, individuals with SUDs report a sense of disappointment resulting from failures in the process of abstinence, social exclusion and social stigma [78, 79]. It is common for people to turn to spirituality in the most difficult life situations, whether driven by loneliness or the feelings of

bottoming out. Studies in this population conclude that the concept of religious teachings and a positive religious context in the family nucleus play a crucial role in the onset and maintenance of recovery [80].

4.5. Rural Populations

Recent shift from urban to rural areas to study the use of crack cocaine and powder cocaine [81, 82] has demonstrated that cocaine availability has increased in rural areas in the South of United States [83]. As access to formal treatment services remains scarce, the population ends up using more of their religious and spiritual approaches for dealing with substance abuse problems [84].

CONCLUSION

Substance use disorders remain an important public health problem with a multifactorial etiology. Unfortunately, there are few effective treatment options. Within this context, spiritual-based approaches emerge as an interesting option. Aggressiveness, which is a very prevalent symptom in individuals with SUD, appears to be particularly favored by a higher level of spirituality. More than a claimed direct effect from god or a 'higher power', these approaches seem to manage some pathological behaviors, improving self-control for recovery. Such interventions can have good effects in reducing some core symptoms associated with substance use, such as aggressiveness. However, spirituality may not play a beneficial role in some subgroups, such as among individuals with crack and cannabis use disorders. The most available type of intervention for alcohol use disorders is a spiritual-based one (AA) which can help the individual to achieve recovery by establishing social networks, and through developing self-efficacy and coping skills. Spirituality seems to be especially beneficial for minorities such as Latino, African-American and American-Indian populations. Their traditional cultures, usually strongly linked to concepts of spiritual or religious basis, can generate a great support for people trying to recover from SUDs, even in the worst environments. Spiritual-based interventions are also good alternatives in rural environments, where professional health treatment for SUDs is not available. However, spiritual-based intervention should not replace the conventional health treatment for SUDs. Probably, a combination of these treatments can be a beneficial alternative for minorities and those with a religious background. There is a need for prospective studies outside U.S., especially where spiritual-based approaches are available. It may be difficult to carry out randomized controlled trials because of the nature of the spiritual/religious dimensions. However, prospective studies that evaluate mediation effect of spirituality and religiosity on recovery would be helpful. Qualitative studies combined with quantitative design offer excellent options to evaluate the recovery process, especially among special populations.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

Dr. Castaldelli-Maia has been awarded with a Pfizer Independent Grant for Learning and Change (IGLC) managed

by Global Bridges (Healthcare Alliance for Tobacco Dependence Treatment) hosted at Mayo Clinic, to support free smoking cessation treatment training in addiction/mental health care units in Brazil (grant IGLC 13513957) and Portugal (grant IGLC 25629313), which had no relationship with the present study. Dr. Andrade is Executive President of Center for Information on Health and Alcohol (CISA), which had no funding relationship with this project. All other authors have no conflicts of interest.

ACKNOWLEDGEMENTS

Declared none.

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